# Wave Reflection and Transmission at Normal Incidence – Part 2

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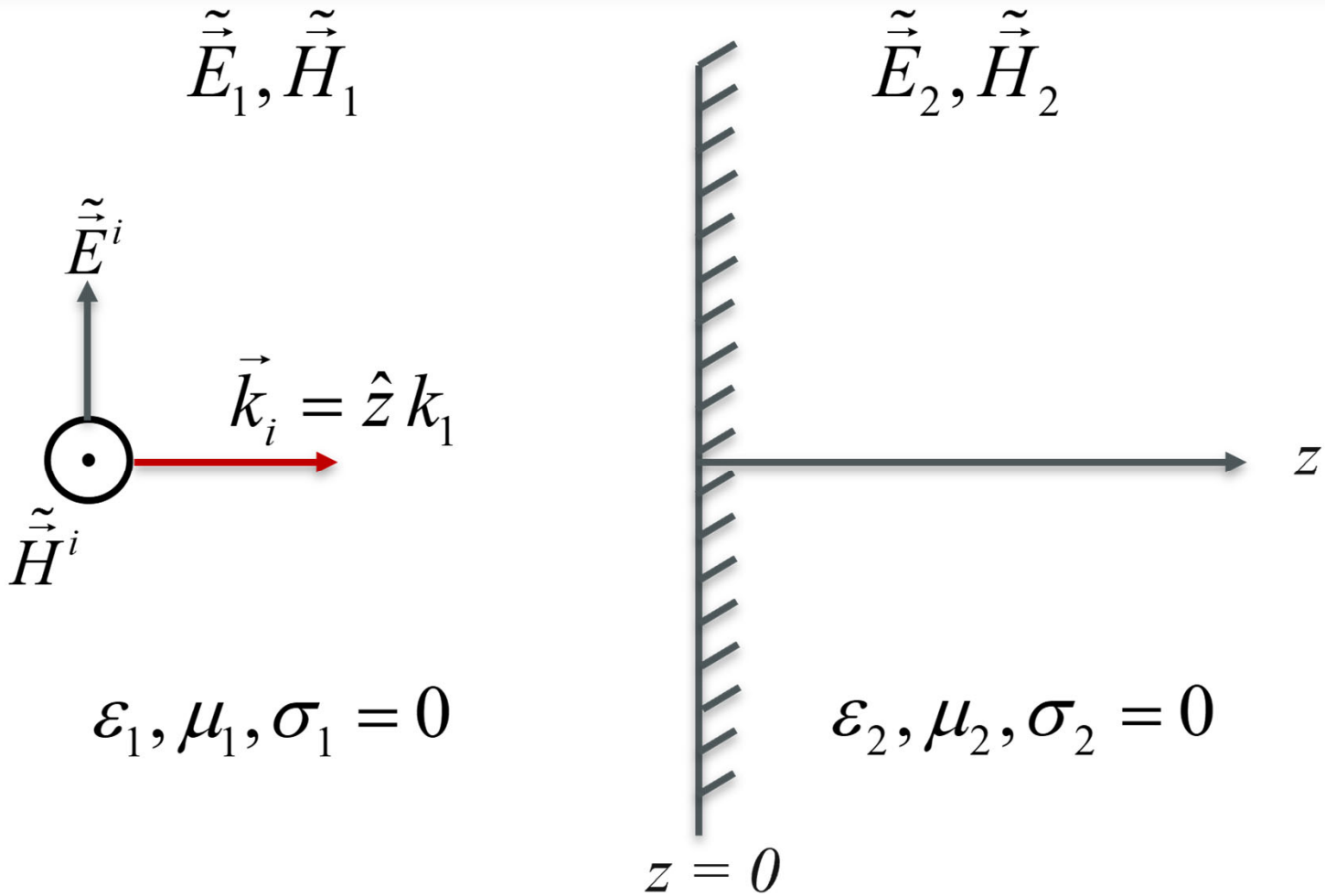
# Lecture Objective

By the end of this lecture, you will be prepared to:

- Analyze plane wave reflection from the boundary between two dielectric half-spaces at normal incidence.

# Normal Incidence On a Plane Dielectric Boundary



# Incident Plane Wave

- In the absence of the obstacle (medium 2), assume an ***incident*** field is established by the source and is given by

$$\tilde{\vec{E}}^i(z) = \hat{x} E_0^i e^{-jk_1 z}$$

- And the corresponding magnetic field is given by

$$\tilde{\vec{H}}^i(z) = \frac{1}{\eta_1} \hat{z} \times \tilde{\vec{E}}^i = \hat{y} \frac{E_0^i}{\eta_1} e^{-jk_1 z}$$

- Where $k_1 = \omega\sqrt{\mu_1 \varepsilon_1}$ $\qquad \eta_1 = \sqrt{\frac{\mu_1}{\varepsilon_1}}$

# Total Field in Regions 1 and 2

- With the obstacle present, the fields in regions 1 and 2 must satisfy both Maxwell's equations (or equivalently, the Helmholtz equation) and the boundary conditions at $z = 0$.
- Since the source excites the incident plane wave in the absence of the obstacle, and the boundary is an infinite plane, it makes sense to assume that the fields in both regions are superpositions of plane waves.

# Total Field in Region 1

- Assume that the total field in region 1 is equal to the incident wave plus a "reflected" wave:

$$\tilde{\vec{E}}_1(z) = \tilde{\vec{E}}^i(z) + \tilde{\vec{E}}^r(z)$$

$$\tilde{\vec{H}}_1(z) = \tilde{\vec{H}}^i(z) + \tilde{\vec{H}}^r(z)$$

- Where

$$\tilde{\vec{E}}^r(z) = \hat{x} E_0^r e^{+jk_1 z}$$

$$\tilde{\vec{H}}^r(z) = -\hat{y} \frac{E_0^r}{\eta_1} e^{+jk_1 z}$$

- The reflected wave satisfies the Helmholtz equation in region 1.
- Physically, the reflected wave is propagating in the –z-direction (back toward source).

$$\vec{k}_r = -\hat{z} k_1$$

# Total Field in Region 2

- In region 2, it makes sense to assume that some of the incident wave will transmit across the boundary and continue propagating in region 2 away from the boundary.
- Assume that the total field in region 2 is this "transmitted" wave:

$$\tilde{\vec{E}}_2(z) = \tilde{\vec{E}}^t(z)$$

$$\tilde{\vec{H}}_2(z) = \tilde{\vec{H}}^t(z)$$

- The transmitted wave satisfies the Helmholtz equation in region 2.

$$k_2 = \omega\sqrt{\mu_2\varepsilon_2}$$

- Where

$$\tilde{\vec{E}}^t(z) = \hat{x}E_0^t e^{-jk_2 z}$$

$$\tilde{\vec{H}}^t(z) = \hat{y}\frac{E_0^t}{\eta_2}e^{-jk_2 z}$$

$$\eta_2 = \sqrt{\frac{\mu_2}{\varepsilon_2}}$$

# Reflected and Transmitted Plane Waves

- As we have postulated them, the reflected and transmitted plane waves are completely specified once $E_0{}^r$ and $E_0{}^t$ are determined.
- To solve for these two unknowns, we need two equations.
- The two equations are obtained by satisfying the boundary conditions on tangential components of electric and magnetic field at $z = 0$.

# Applying BCs

- Continuity of tangential electric field:

$$\tilde{\vec{E}}_1(z=0) = \tilde{\vec{E}}_2(z=0) \quad \Rightarrow \quad E_0^i + E_0^r = E_0^t$$

- Continuity of tangential magnetic field:

$$\tilde{\vec{H}}_1(z=0) = \tilde{\vec{H}}_2(z=0) \quad \Rightarrow \quad \frac{E_0^i}{\eta_1} - \frac{E_0^r}{\eta_1} = \frac{E_0^t}{\eta_2}$$

# Reflection Coefficient

- Solving for $E_{r0}$:

$$E_{r0} = \Gamma E_{i0}$$

- Where

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$$

**Reflection coefficient**

# Transmission Coefficient

- Solving for $E_{t0}$:

$$E_{t0} = \tau E_{i0}$$

- Where

$$\tau = \frac{2\eta_2}{\eta_2 + \eta_1}$$

**Transmission coefficient**

# Reflection and Transmission Coefficients

- $\Gamma$ and $\tau$ are related by: $\tau = 1 + \Gamma$
- Notice that

| $\eta_2 / \eta_1$ | $\Gamma$ | $\tau$ |
|---|---|---|
| $<<1$ | $\rightarrow -1$ | $\rightarrow 0$ |
| $1$ | $0$ | $1$ |
| $>>1$ | $\rightarrow 1$ | $\rightarrow 2$ |

$$-1 \leq \Gamma \leq 1$$
$$0 \leq \tau \leq 2$$

# Question 1

- Assume a plane wave traveling in free space is normally incident on the boundary with a non-magnetic lossless dielectric half space half-space having a dielectric constant of 4. Evaluate the reflection and transmission coefficients.

# Question 2

- What happens to the reflection and transmission coefficients when one or both of the media are lossy?
- Assume a plane wave traveling in free space is normally incident on the boundary with a non-magnetic lossless dielectric half space half-space having a dielectric constant of 4 and a dissipation factor of 0.02. Evaluate the reflection and transmission coefficients.

# Total Field in Region 1 Continued

- The total electric field in region 1 is given by:

$$\tilde{\vec{E}}_1(z) = \tilde{\vec{E}}^i(z) + \tilde{\vec{E}}^r(z) = \hat{x}E_0^i\left(e^{-jk_1z} + \Gamma e^{+jk_1z}\right)$$

$$= \hat{x}E_0^i e^{-jk_1z}\left(1 + \Gamma e^{j2k_1z}\right)$$

- The total magnetic field in region 1 is given by:

$$\tilde{\vec{H}}_1(z) = \tilde{\vec{H}}^i(z) + \tilde{\vec{H}}^r(z) = \hat{y}\frac{E_o^i}{\eta_1}\left(e^{-jk_1z} - \Gamma e^{+jk_1z}\right)$$

$$= \hat{y}\frac{E_{i0}}{\eta_1}e^{-jk_1z}\left(1 - \Gamma e^{j2k_1z}\right)$$

# Total Field in Region 2 Continued

- The total electric field in region 2 is given by:

$$\tilde{\vec{E}}_2(z) = \tilde{\vec{E}}^t(z) = \hat{x}\tau E_0^i e^{-jk_2 z}$$

- The total magnetic field in region 2 is given by:

$$\tilde{\vec{H}}_2(z) = \tilde{\vec{H}}^t(z) = \hat{y}\tau \frac{E_0^i}{\eta_2} e^{-jk_2 z}$$

# Total Time-Domain Field in Region 1

- The total time-domain electric field in region 1 is given by:

$$\tilde{\vec{E}}_1(z,t) = \mathrm{Re}\left\{\hat{x}E_0^i\left(e^{-jk_1z} + \Gamma e^{+jk_1z}\right)e^{j\omega t}\right\}$$

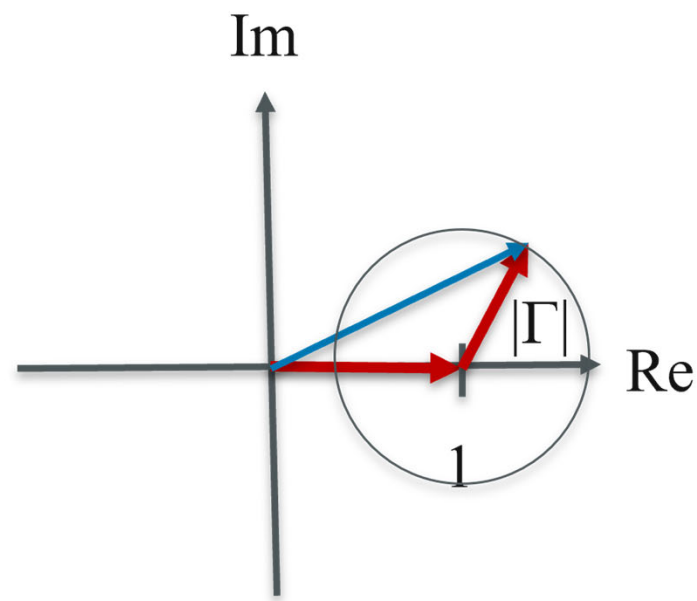
$$= \mathrm{Re}\left\{\hat{x}E_0^i\left[\Gamma\left(e^{-jk_1z} + e^{+jk_1z}\right) + (1-\Gamma)e^{-jk_1z}\right]e^{j\omega t}\right\}$$

$$= \hat{x}E_0^i\left[2\Gamma\cos(k_1z)\cos(\omega t) + (1-\Gamma)\cos(\omega t - k_1z)\right]$$

"Standing wave" "Traveling wave"

# Standing Wave Pattern for the Electric Field in Region 1

$$\left|\tilde{\vec{E}}_1(z)\right| = E_0^i\left|1 + \Gamma e^{+j2k_1 z}\right|$$



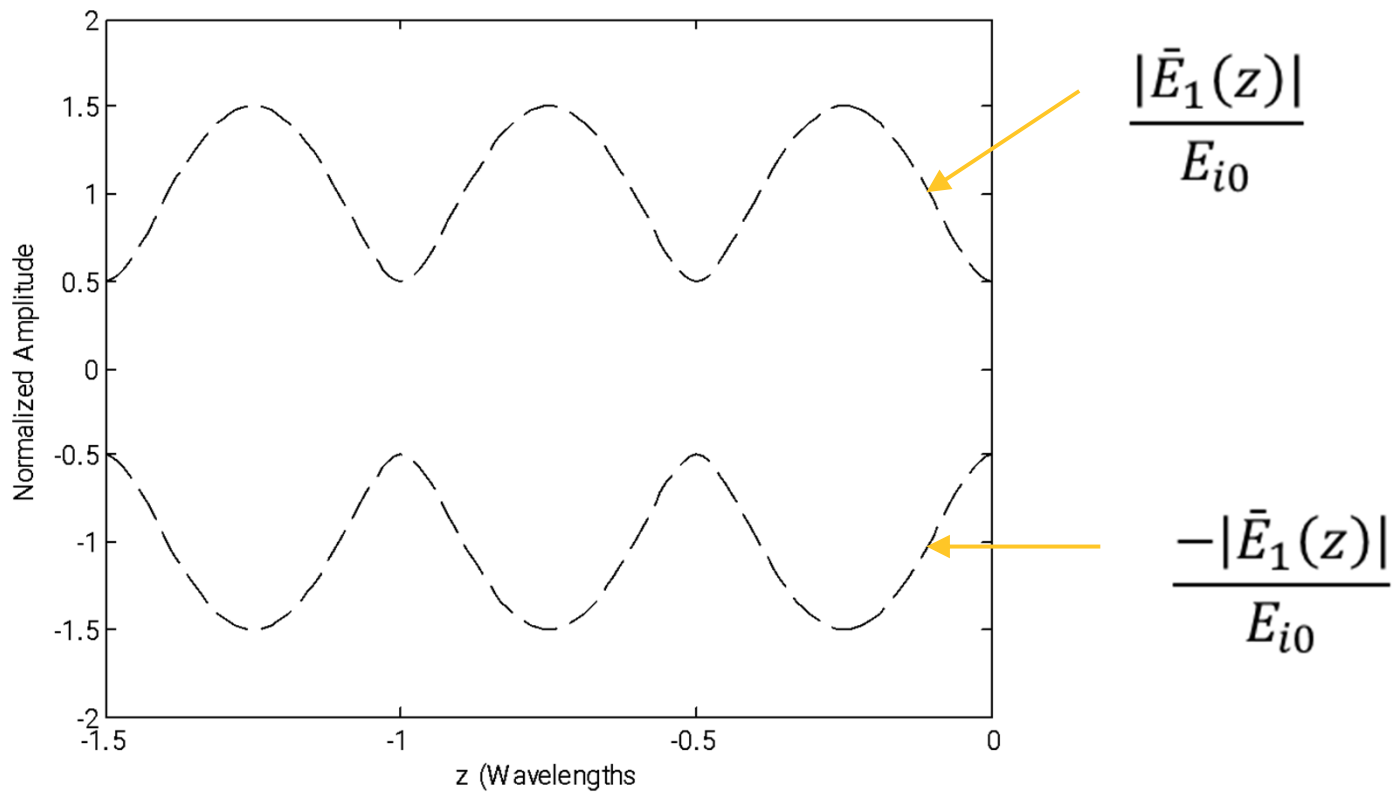

$$\left|\tilde{\vec{E}}_1(z)\right|_{\max} = E_0^i\left(1+|\Gamma|\right)$$

$$\left|\tilde{\vec{E}}_1(z)\right|_{\min} = E_0^i\left(1-|\Gamma|\right)$$

Periodicity of function:

$$2k_1\Delta z = 2\pi \rightarrow \Delta z = \frac{\lambda_1}{2}$$

# Standing Wave Pattern for the Electric Field



# Summary

- A plane wave normally incident on the planar boundary between two dielectric half-spaces is partially reflected back into the first medium and partially transmitted into the second medium.
- The reflection and transmission coefficients quantify how much reflection and transmission occurs.
- In the first medium, the total field can be written as the sum of a standing wave plus a traveling wave.

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